

Pressure & vacuum monitoring comes with unique safety, compliance and maintenance demands. That's why **Unique Vacuum Solutions** wants to ensure that you have expertly calibrated equipment. At our works, we perform vacuum gauge calibrations to support critical objectives – from safeguarding product stability to ensuring research validity. Our talented experts have the knowledge to conduct ultra-accurate calibrations so your process vacuum equipment delivers the precision you need.

## **Calibrate Vacuum Gauges for High Performance**

- Ensure production quality or pinpoint issues in production.
- Receive correct gauge readings that contribute to uptime and safety.
- Keep equipment certifications up to date in regulated environments.
- Depend on accurate instruments during monitoring or testing.
- Prepare newly purchased devices for duty and verify application compatibility.
- Identify and correct equipment errors – before services and processes are impacted.

## **Why Choose Unique Vacuum Solutions Calibration?**

Our extensive [vacuum calibration services](#) include evaluating devices that report vacuum, absolute and gauge measurements. Zero and span adjustments where ever necessary in consultation with customer. We calibrate all types of vacuum measurement instrument, including [digital and analog vacuum gauge calibrations](#) :

- Capacitance manometers from full scale range 1 torr and 1 mbar.
- Pirani vacuum measurement gauges: range 10e-4 mbar to 20 mbar.
- Cold cathode vacuum gauges: 10e-2 mbar to 10e-6 mbar range.
- Hand held battery operated refrigeration vacuum gauges.

Unique vacuum solutions will work with you to establish a schedule for each vacuum measurement device that requires recalibration. We can then provide timely vacuum calibration notifications. You can also rely on rapid turnarounds, with many devices back on the job in just five business days.

## **Job execution:**

Unique Vacuum Solutions has trained and expert calibration staff who carry the master reference vacuum gauge along with the high vacuum generation system (turbo pump based) to the customer site.

## ***The master calibration references:***

- for medium vacuum range up to 1 micro bar or 0.001 mbar will be capacitance manometer BARATRON
- for high vacuum measurement from 0.01 mbar to 10e-6 mbar the master will be the full range gauges.
- All the masters will have the valid NPL or NIST traceable certificates.

## **Lead time to attend the calibration:**

Lead time to attend the calibration from the date of receipt of confirmed order: 04 days for the metro and major cities, 8 working days for the non metro and tier II cities. The calibration procedure and master calibration reference certificates will be submitted prior to the commencement of the calibration.

## The procedure:

The INSTRUMENT UNDER CALIBRATION (IUC) is assembled along the master gauge, needle vacuum valve

To ensure the ultimate base pressure of 10-5mbar pre-evacuation, we shall be carrying one set of high vacuum gauge set for measuring the 10e-5 mbar during calibration proceedings.

- 1) *The required reference instruments, vacuum generation system which consists of turbo molecular vacuum pumping system with all vacuum connections, calibration dome and fittings will be brought by UVS engineer.*
- 2) *We shall submit the calibration report of the vacuum gauge upon completion of the calibration.*
- 3) *Calibration range for vacuum gauge: 10mbar to 10e-3 mbar. OR 10000 micro bar to 1 micro bar.*

## Actual photos taken during on site calibration:







Vacuum Products & Helium Leak Test

## UNIQUE VACUUM SOLUTIONS

| CALIBRATION PROCEDURE FOR VACUUM GAUGES          |                                                                                                  |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Doc. No.                                         | Date : 15.08.16                                                                                  |
| UVS/OS/CAL/CP/RC/014/REV1.0                      |                                                                                                  |
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### OBJECTIVE

: This Procedure is established to calibrate the Vacuum Gauges with the Help of Calibration Unit.

### SCOPE

: This Procedure Shall be applied to Calibrate the Vacuum Gauges/Vacuum Transmitters with or without Display controllers.

### METHODOLOGY

: Calibration of Vacuum Gauges/Transmitters by comparing with the calibrated Master Gauge Reading traceable to NPL or NIST

### Standard Equipments Used:

- Calibration Unit with Standard Ports consisting of TMP with rotary vane vacuum pumping system with valves for roughing, backing and gas purging.
- Vacuum Transmitters & Display Units under calibration.
- Master vacuum gauges with the display units.
- Standard DC Power Supply in case the transmitters needs external power supply.
- Multi meter for measuring the output voltage in case of no vacuum readout options in the gauge under calibration.

### Procedure :

1. Connecting the Master Gauges(Calibrated) and Gauges for the Calibration in the respective ports on the Calibration Chamber.
2. Evacuating the Calibration Chamber to a Vacuum level of 10(-4) m.bar or better using the turbo molecular pump.
3. **initial base pressure stabilisation:** The Evacuation process will continue for about minimum of 30 minutes for the stabilization of the Master gauges and the gauges to be calibrated or once the stability in the both the master and the under calibration gauge readings are stable.- which ever is earlier.
4. take the measurement at different points by controlling the vacuum with the help of fine control needle valve depending upon the Vacuum range of the Gauge.

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5. **observing and noting the values** : the intermediate values from 1 micro bar or the minimum value of the master till the full scale value of the master, there will be minimum of 4 intervals readings to be taken.

6. holding time for intermediate values during recording the timing:

As the total range is under vacuum, the control of air bleeding from micro control needle valve and the high vacuum evacuation valve is managed and controlled in such a pattern, where the value from 1 micro bar (0.001 mbar) to 1000 micro bar ( 1 mbar) is stopped at minimum of 4 intervals for a minimum duration of 1 second or till the time where both the readings are seen simultaneously before the change in any one of the reading.

6. adjusting the zero and full scale in the under calibration instrument:

For adjusting the zero, evacuate the sensor to minimum vacuum level one order better than the resolution of the under calibration instrument or as per the OEM.

For adjusting the full scale, simulate the pressure in the calibration manifold to the full scale value of the master and adjust the full scale in the under calibration instrument accordingly.

Repeat the above zero and full scale adjustment procedure if needed and check for the intermediate values.

- **For pirani or Baratrons with range from 0.001 mbar and above.**

The readings are taken at maximum of eight intervals:

1. 0.001 mbar to 0.005 mbar : 01 reading
2. 0.005 mbar to 0.020 mbar : 01 reading
3. 0.020 mbar to 0.100 mbar : 01 reading
4. 0.100 mbar to 0.250 mbar : 01 reading
5. 0.250 mbar to 1.0 mbar : 01 reading

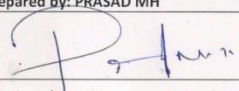
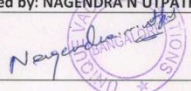
- **For high vacuum gauges (cold cathode, BA) measurement up to 10e-6 mbar:**

The readings are taken at maximum of seven intervals:

1e-6, 4e-6, 2e-5, 7e-5, 2e-4, 7e-4, 1e-3 mbar ranges.

5. record Master Gauge reading and corresponding under calibration gauge reading at each measured point.

6. provide the Calibration report with all the details of the Gauge, recorded reading, master gauge used with calibration certificate number details.

|                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
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